AMIA/HL7 FHIR App Competition 2026 - Student Application Draft

CLARA-Care: Governed Longitudinal Health Context for Persistent AI Workflows

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1. Title

CLARA-Care: Governed Longitudinal Health Context for Persistent AI Workflows

2. Category

Student

3. Letter of Support

Required before submission. Use the included primary-advisor attestation template; do not upload an unsigned placeholder.

4. Project abstract (max 1,000 characters)

CLARA-Care is a research prototype for longitudinal personal-health AI that unifies model-visible context and persistent updates into one traceable, governed workflow. Integrating SMART-on-FHIR authorization and HL7 FHIR R4 Provenance preservation, it ingests longitudinal FHIR Bundles into a source-preserving timeline, compiles a Task-Bounded Health State Snapshot (THSS) parameterized by granted SMART scopes, and routes model proposals through Governed State Transitions (GST). Unlike conventional FHIR REST ETag (If-Match) optimistic locking—which only verifies single-resource version payloads and remains blind to asynchronous dynamic consent revocation, policy epoch drift, and cross-domain atomicity—CLARA-Care re-evaluates base versions, delegated SMART authorization, and clinical safety invariants at commit time. Evaluation on synthetic and benchmark cohorts demonstrates robust TOCTOU prevention and audit reconstruction without direct unvetted EHR writeback.

5. Project rationale, impact and innovation (max 3,500 characters)

Longitudinal health assistants repeatedly read from and propose updates to evolving clinical records. However, relevant retrieval alone does not guarantee that an AI sees an accurate current state, that disclosed evidence adheres to current authorization, or that a generated proposal remains admissible after clinical or governance conditions shift. Standard HL7 FHIR REST architectures address concurrency via HTTP ETag (If-Match) optimistic locking on individual resource instances. While effective for synchronous single-resource edits, ETag mechanisms exhibit critical vulnerabilities in persistent, multi-turn AI workflows: 1. Multi-Resource Inconsistency: Complex clinical workflows span multiple resources (e.g., updating Conditions, MedicationRequests, and Observations simultaneously); independent per-resource ETag checks allow partial commits and cross-domain state corruption. 2. Governance-TOCTOU Blindness: HTTP ETags track only entity payload revisions. They cannot detect out-of-band governance drift—such as dynamic consent revocation, role boundary changes, or institutional policy epoch transitions—occurring during the multi-second latency of LLM inference. 3. False-Stale Contention: Coarse container-level ETags trigger excessive aborts when concurrent agents update mutually disjoint health domains.

CLARA-Care overcomes these limitations by decoupling canonical evidence from derived model context and linking inference context to Governed State Transitions (GST). The architecture incorporates three foundational standards-based innovations: 1. SMART-on-FHIR Authorization Integration: OAuth 2.0 / OpenID Connect launch context and fine-grained SMART v2 scopes (e.g., patient/*.rs, user/*.cruds) directly parameterize the compilation of Task-Bounded Health State Snapshots (THSS), ensuring model-visible context is strictly confined to authorized clinical data. At writeback, GST re-evaluates caller delegated authority at commit time, preventing privilege escalation. 2. HL7 FHIR R4 Provenance Preservation: The system ingests STU3/R4 resources while retaining complete source payloads, version identifiers, and SHA-256 hashes. It synthesizes standards-compliant FHIR R4 Provenance resources (recording target, recorded, agent, and entity derivation lineage), guaranteeing immutable audit trails and snapshot reproducibility across derived-state rebuilds. 3. Dual-Layer Governed Writeback vs. REST ETags: Rather than relying on uncoordinated ETag updates or direct record writes, AI proposals undergo two-layer validation that re-checks base state freshness, active consent grants, dynamic policy epochs, and deterministic clinical safety constraints (e.g., DDI barriers) before persistence.

CLARA-Care uses FHIR as a standardized interoperability and provenance foundation. It demonstrates how FHIR-derived events can safely drive governed AI workflows without direct writeback, making no claim of clinical deployment or regulatory certification.

6. Project design and implementation (max 7,000 characters)

The prototype is organized around a longitudinal state layer with explicit read and write boundaries, separating canonical evidence, task-bounded AI context, and a governed commitment gateway.

1. Source-Preserving Ingestion and HL7 FHIR R4 Provenance Preservation The version-explicit ingestion module processes FHIR STU3 and R4 Bundles (collection or transaction). It enforces strict subject partitioning (requiring exactly one Patient per Bundle and validating all subject references) to prevent cross-patient contamination. The system ingests canonical resources into an append-only longitudinal timeline while preserving the raw source resource JSON payload, resource identity, version ID, and cryptographic SHA-256 digest.

Coded concepts (SNOMED CT, RxNorm, LOINC, ICD-10) and encounter references are extracted while strictly distinguishing clinical valid/effective time (effectiveDateTime/period) from ingestion known-at time (recorded/authoredOn). Supported resources include Patient, AllergyIntolerance, CarePlan, Condition, MedicationRequest, Observation, Procedure, Provenance, and version-specific ServiceRequest (R4) or ProcedureRequest (STU3).

To guarantee end-to-end auditability, CLARA-Care synthesizes and maintains standards-compliant HL7 FHIR R4 Provenance records. Each timeline event and derived state artifact links target resources to their originating entities (Provenance.entity.role = derivation or source, Provenance.entity.what pointing to the canonical resource hash) and participating agents (Provenance.agent specifying practitioner, patient, or automated AI device with corresponding provenance roles). Provenance closure is preserved across full rebuilds of derived longitudinal views.

2. SMART-on-FHIR Authorization Integration CLARA-Care integrates standard SMART-on-FHIR (OAuth 2.0 / OpenID Connect) authorization profiles. During session establishment, launch parameters (patient context, encounter context) and fine-grained SMART v2 scopes (e.g., patient/Observation.rs?category=vital-signs, patient/MedicationRequest.r, user/*.cruds) are extracted and bound to the request.

For AI inference, CLARA-Care compiles a Task-Bounded Health State Snapshot (THSS). THSS synthesis is parameterized by the authenticated actor, clinical purpose of use, task definition, and active SMART scopes, filtering longitudinal records so that the model context contains exclusively authorized, task-relevant clinical facts. Dynamic patient consent directives (modeled via FHIR Consent states) are evaluated during snapshot generation to enforce granular disclosure constraints.

3. Governed State Transitions (GST) vs. FHIR REST ETag Optimistic Locking In standard HL7 FHIR REST architectures, concurrent mutations rely on HTTP optimistic locking via ETag and If-Match headers (e.g., If-Match: W/2̈). While ETag prevents lost updates on isolated single-resource endpoints, it presents critical deficiencies in stateful, multi-turn AI workflows: - Lack of Cross-Domain Atomicity: Multi-resource AI proposals (e.g., updating a Condition while concurrently modifying MedicationRequests and recording Observations) execute as separate HTTP requests, risking partial commits and inconsistent clinical states. - Inability to Detect Governance Drift (TOCTOU): HTTP ETags inspect only resource payload versions; they cannot detect out-of-band dynamic consent revocations, role expirations, or policy epoch updates occurring while an LLM generates a response over a multi-second window. - False-Stale Aborts: Coarse container-level ETags reject concurrent non-conflicting updates to disjoint health domains.

CLARA-Care replaces direct FHIR REST updates with Governed State Transitions (GST). Model outputs cannot write to canonical storage directly. When an AI generates a durable change proposal, it returns a candidate transition bound to the THSS snapshot hash. At commit time, GST executes a multi-point verification barrier: a. Base-State Freshness: Verifies that underlying resource versions have not suffered conflicting modifications. b. Commit-Time Authority Verification: Re-evaluates active SMART-on-FHIR scopes and dynamic consent policies to guarantee authority remains valid at the exact commitment instant, eliminating Time-of-Check to Time-of-Use (TOCTOU) vulnerabilities. c. Deterministic Clinical Safety: Evaluates hard clinical invariants (e.g., severe drug-drug interaction screening and contraindicated allergy rules). d. Provenance Ledger Emission: Upon successful validation, persists the update along with an immutable FHIR R4 Provenance record linking the commit to the originating THSS snapshot, governing policy epoch, and authorizing clinician agent.

The system is a research prototype evaluating governed AI interoperability; it does not claim production EHR deployment or CDS Hooks integration without dedicated verification.

7. Project evaluation and sustainability (max 3,500 characters)

Evaluation rigorously benchmarks interoperability ingestion, model-context task fidelity, and transactional governance safety under concurrency against alternative paradigms.

1. Interoperability and Ingestion Scale: Local ingestion tests verified 1,307,771 SyntheticMass FHIR Bundles; 927,109 MIMIC-IV Demo FHIR records for 100 subjects; and 540,237 normalized eICU Demo records covering 1,841 subjects and 2,520 ICU stays. In all cases, 100

2. Model-Visible Context Utility: In a frozen 64-subject synthetic cohort, Strict THSS compilation guided by SMART scope boundaries achieved all-axis exact match for 63/64 subjects with Claude Sonnet 4.6 and 64/64 with Gemini 3.6 Flash High. A source-disjoint six-task sanity grid completed 48/48 calls, achieving 11/12 correct Strict-THSS cells versus 9/12 under unbound context. A sealed 384-subject Claude comparison between Strict THSS and full authorized history showed null difference, confirming that bounded scoping minimizes token disclosure without degrading reasoning accuracy.

3. Comparative Concurrency and Governance Safety vs. FHIR REST ETags: In a 16-worker PostgreSQL 16 comparative concurrency benchmark testing 500 clinical transactions across 5 scenario families (single entity, cross-domain multi-resource, dynamic consent revocation TOCTOU, disjoint parallel slots, and severe DDI conflicts): - Standard FHIR REST Optimistic Locking (ETag / If-Match) exhibited a 75.0- MemTX (stateful agent memory) exhibited 22.0- CommitGuard achieved 0.0- CLARA-Care Governed State Transitions achieved 0.0

Sustainability is ensured through an open, modular repository, deterministic state hashing, immutable provenance graphs, and clear separation of canonical state from derived views.

8. Intended users / audience

Researchers and developers building longitudinal personal-health assistants, patient-facing health-record tools, or health-informatics prototypes that need interoperable data ingestion, SMART-on-FHIR authorization, and explicit governance of AI context and persistent writeback.

9. 140-character project summary

CLARA-Care links FHIR R4 Provenance and SMART auth to task-bounded AI context and governed write transitions beyond REST ETag locks.

10. How is FHIR used? (max 500 characters)

CLARA-Care ingests FHIR STU3/R4 Bundles into a source-preserving timeline, retaining full HL7 FHIR R4 Provenance metadata and SHA-256 hashes. SMART-on-FHIR authorization scopes constrain Task-Bounded Health State Snapshots (THSS) provided to AI models. Durable writebacks are governed by Governed State Transitions (GST) rather than bare FHIR REST ETag updates, re-validating base versions, active consent grants, and delegated SMART scopes at commit time to eliminate TOCTOU vulnerabilities.

11. FHIR release and resources (max 500 characters)

Supports FHIR STU3 and R4 Bundles. Common resources: Patient, AllergyIntolerance, CarePlan, Condition, MedicationRequest, Observation, Procedure, and Provenance. R4 additionally supports ServiceRequest; STU3 supports ProcedureRequest. The system preserves canonical resource payloads, version IDs, and cryptographic hashes, generating conformant FHIR R4 Provenance records linking targets, agents, and derivation entities for complete audit reproducibility.

12. FHIR data source and access (max 500 characters)

Inputs include SyntheticMass FHIR Bundles and MIMIC-IV Demo FHIR datasets, with non-FHIR data normalized to the longitudinal interface. SMART-on-FHIR authorization profiles (OAuth 2.0 / OpenID Connect scopes) define user, patient, and encounter access boundaries. Tested via verified Bundle/file ingestion and local endpoint simulations rather than live production EHR deployments.

13. FHIR resources / Capability Statement

Upload a resource list for the current prototype (included in this pack). Do not upload a CapabilityStatement unless a conformant FHIR server endpoint has actually been implemented and verified.

14. US Core or other implementation guides?

No dedicated US Core conformance claim in the current prototype. The ingestion module accepts supported generic STU3/R4 Bundle resources and models standard FHIR R4 Provenance and SMART authorization scopes.

15. FHIR technologies used

FHIR Bundle ingestion, HL7 FHIR R4 Provenance preservation, and SMART-on-FHIR authorization profile modeling. No current claim for CDS Hooks, Bulk Data, or CQL.

16. Other information (max 1,500 characters)

CLARA-Care is a student-led research prototype investigating governed interoperability for clinical AI. Its design bridges HL7 FHIR standards (STU3/R4 Bundle ingestion, SMART-on-FHIR authorization scoping, and FHIR R4 Provenance resource modeling) with transactional safety barriers that overcome the fundamental concurrency and TOCTOU limitations of conventional FHIR REST ETag optimistic locking.

Supported raw source resources and cryptographic digests are preserved immutably alongside normalized timeline structures, enabling complete audit reconstruction across derived longitudinal views. The commitment gateway guarantees that model proposals derived from Task-Bounded Health State Snapshots cannot silently downgrade to unvetted writes or bypass commit-time governance and safety re-evaluations.

All reported evaluations represent software benchmarks and synthetic cohort experiments. The prototype does not provide medical diagnosis or treatment, is not deployed for patient care, and does not claim regulated medical device certification.

17. Paying customers?

No.

18. When conceived?

2025.

19. When implemented?

Research prototype developed during 2025–2026; update with the exact project milestone date used in your records.

20. Users / patients impacted

No patient-impact claim. Research and development prototype; clinical user count not established.

21. Website / URL

`https://github.com/Project-CLARA-HBT/CLARA-Care`

22. SMART App Gallery publication if feasible

Yes, if the organizers determine the prototype is technically suitable and the published listing accurately describes its current FHIR capabilities.